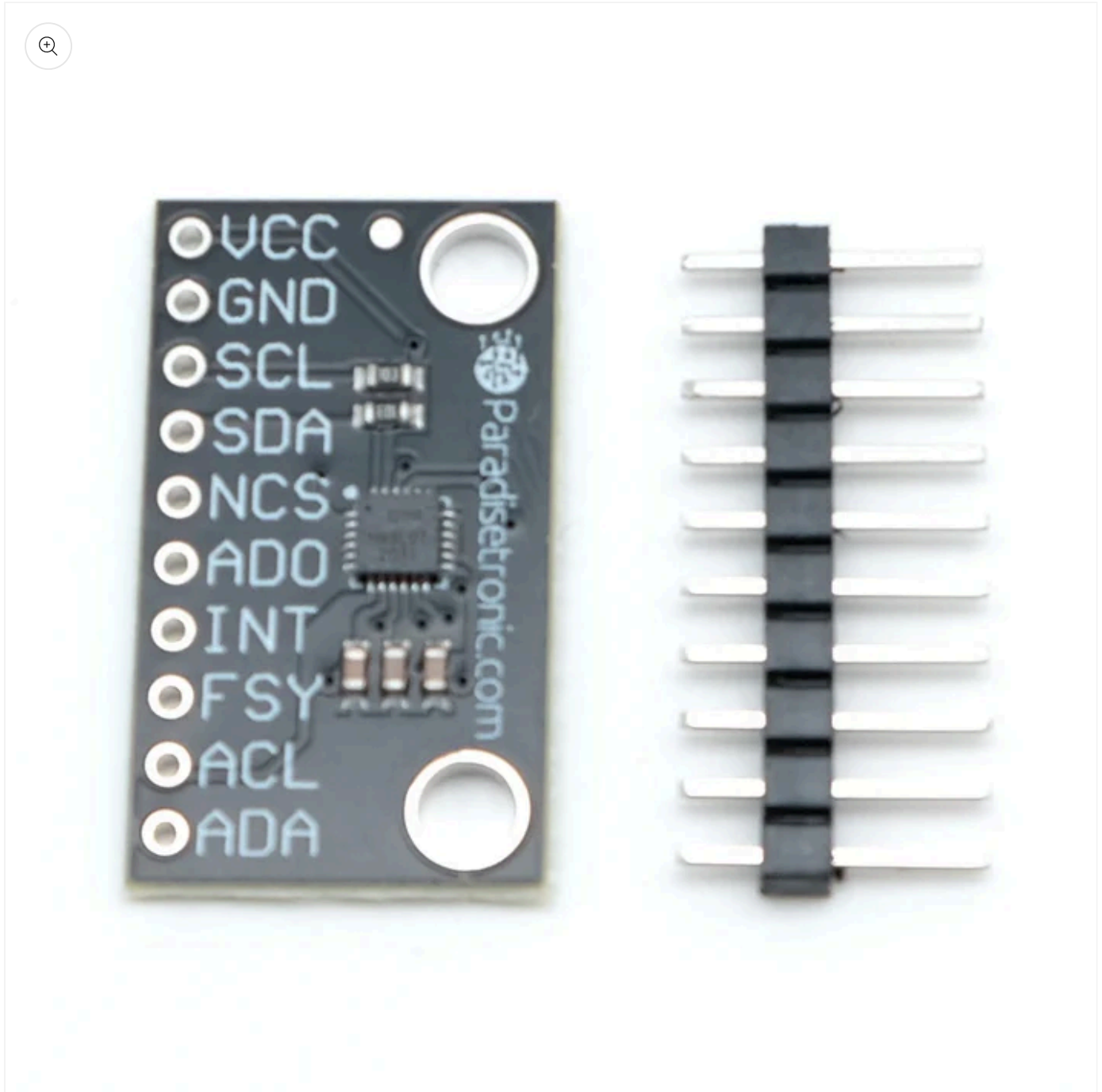
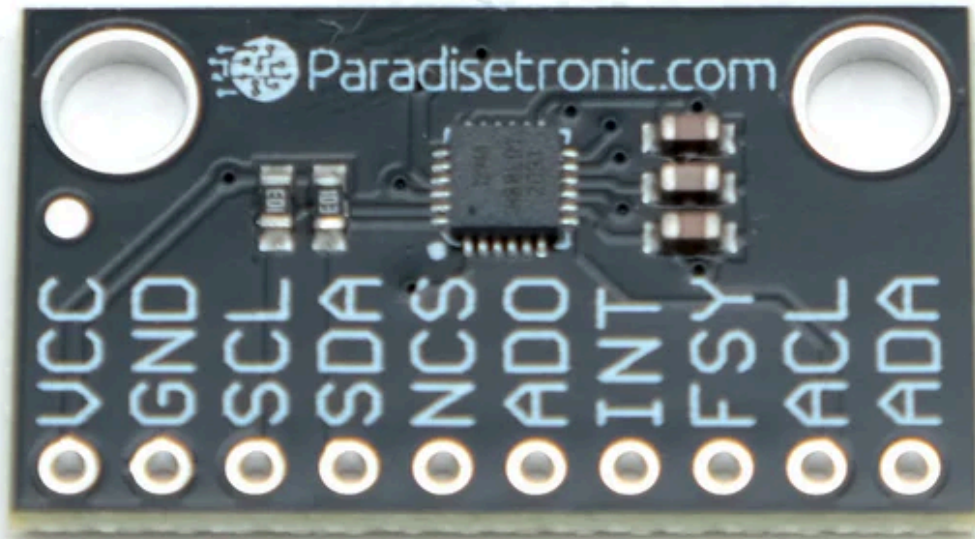
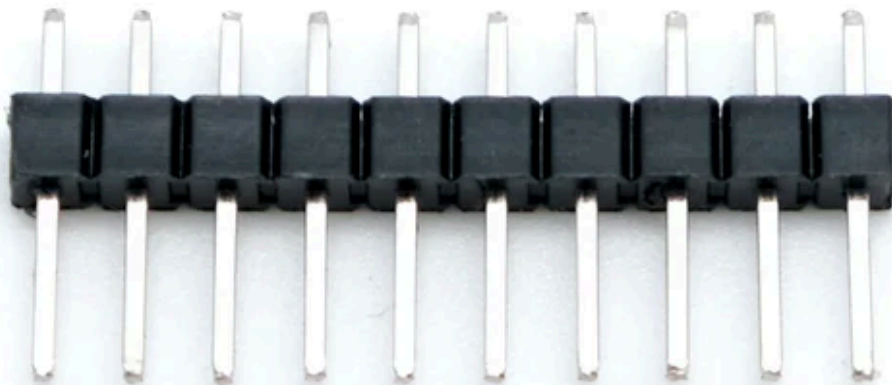
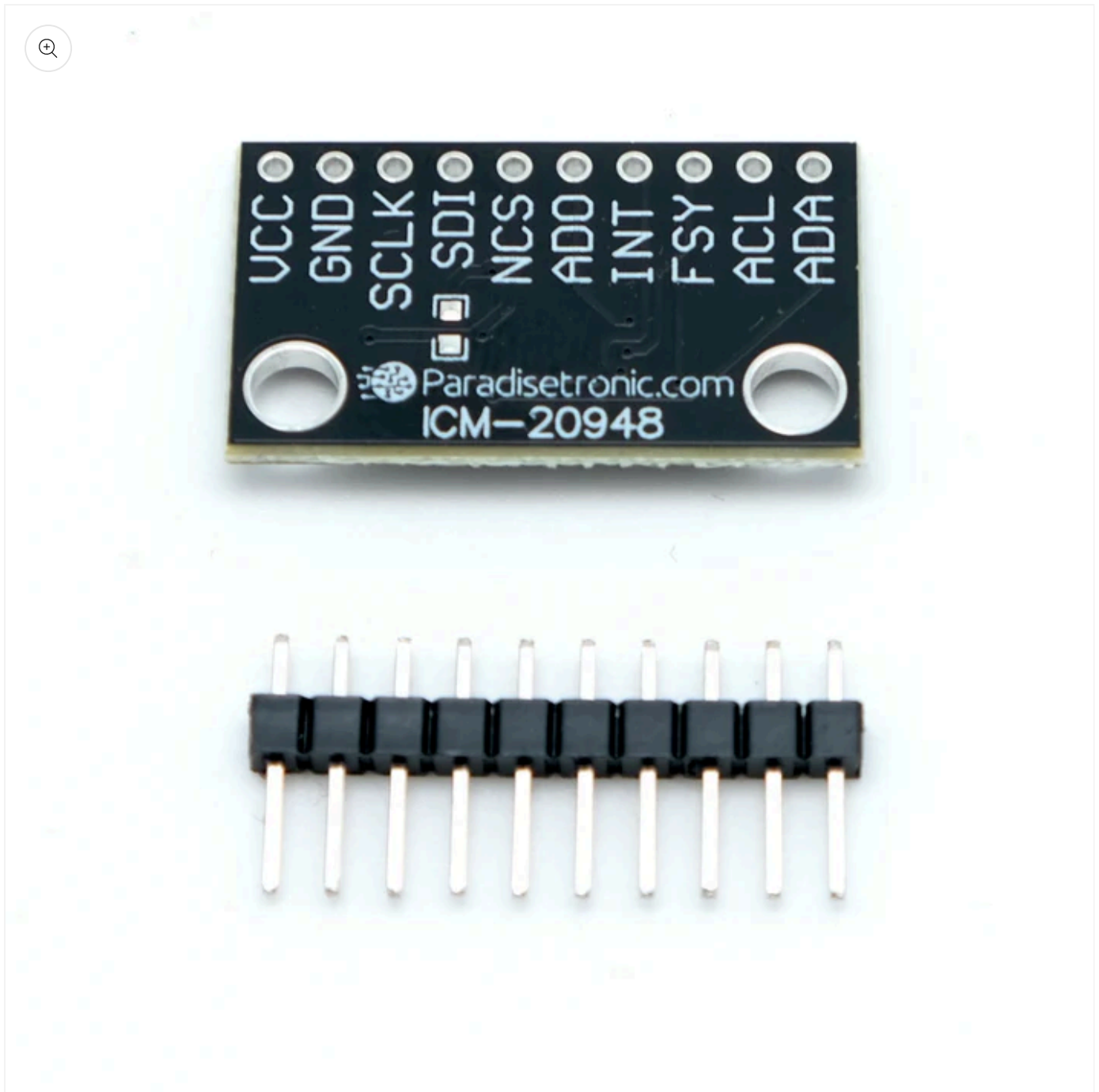


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The ICM-20948 9-Axis Motion Sensor Module is the perfect solution when you need precise motion data for robotics, drones, wearables, or IoT projects. Featuring a 3-axis accelerometer, 3-axis gyroscope, and 3-axis magnetometer in a compact module, it provides accurate orientation and motion detection—all in one! The board is clearly laid out with well-labeled pins, making it easy to integrate into your projects.

Technical Details:

- **9 Degrees of Freedom (9-DoF):** Combines accelerometer, gyroscope, and magnetometer for complete motion tracking.
- **Power-efficient:** Ideal for battery-powered devices thanks to various low-power modes.
- **Highly accurate:** Digital Motion Processing (DMP) delivers precise quaternion and Euler angle data.
- **Wide measurement range:**
 - Accelerometer: $\pm 2g$ to $\pm 16g$
 - Gyroscope: $\pm 250dps$ to $\pm 2000dps$
 - Magnetometer: $\pm 4900\mu T$
- **I2C/SPI Interface:** Easy connection to microcontrollers like Arduino, Raspberry Pi, and others.

Applications:

- Drone stabilization and navigation
- Fitness trackers and smartwatches
- Virtual Reality (VR) or Augmented Reality (AR) applications
- Robotics and autonomous vehicles

- Motion control in IoT devices

Why the ICM-20948?

With built-in 16-bit ADCs, programmable filters, and a temperature sensor for even more accurate results, this module is a true all-rounder. It delivers reliable data even in challenging environments, thanks to low noise and high sensitivity. The I2C/SPI interfaces and clear pin labelling make integrating it into your projects effortless.

Pin Details:

The board comes with a header (unsoldered), giving you flexible connection options. Here's a detailed overview of the pins and how to use them in your projects:

- **VCC:** Supply the board with a voltage between 1.8V and 3.6V.
- **GND:** Ground connection to be linked with the GND pin of your microcontroller to establish a common ground.
- **SCL/SDA:** Used for the I2C interface.
 - **SCL (Serial Clock):** Provides the clock signal.
 - **SDA (Serial Data):** Transfers the data.
- **NCS:** Chip select pin for SPI mode. If using I2C, this pin can be ignored or pulled HIGH (3.3V) to disable SPI mode.
- **AD0:** Determines the I2C address of the module.
 - LOW (GND): Address is 0x68
 - HIGH (VCC): Address is 0x69

Useful when using multiple I2C devices on the same bus without address conflicts.

- **INT:** Interrupt pin sends a signal when a specific event occurs, e.g., when new data is ready or a motion threshold is exceeded.
- **FSY:** Frame Sync pin allows external synchronization, e.g., for time-critical applications like video recording or precise motion tracking. In most hobby projects, this can be left unconnected.

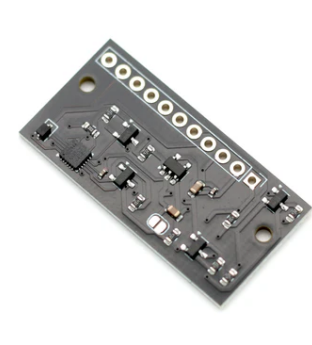
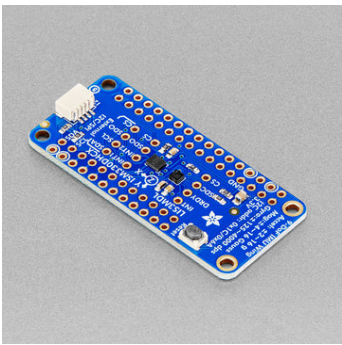
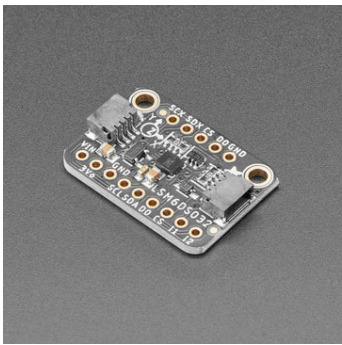
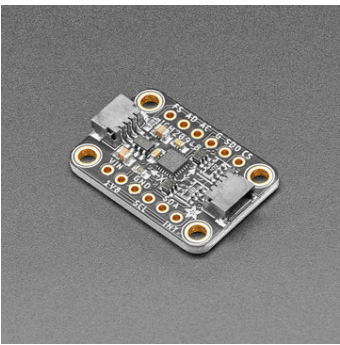
- **ACL/ADA:** These pins are for advanced features such as communication with an external I2C device (e.g., an additional sensor) via the module. If unused, they can be left unconnected.

Note: The module is designed for 3.3V systems; for 5V systems, the use of a logic level shifter is recommended.

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